

Foundations of Dropout regularization

[Dropout regularization]

$$\tilde{h} = r \odot h, \quad r_i \sim \text{Bernoulli}(p)$$

1.0 Content Block

$$R(w) = \sum_{g=1}^G \|w_g\|_2^2, \quad \text{where } \|w_g\|_2^2 = \sum_{j=1}^{|G_g|} (w_{g,j})^2 \quad \left\{ \displaystyle \|w_g\|_2 = \sqrt{\sum_{j=1}^{|G_g|} (w_{g,j})^2} \right\}$$

2.0 Content Block

$$\min_f \sum_{i=1}^n V(f(\hat{x}_i), y_i) + \lambda \|f\|_{\mathcal{H}}^2 \quad \left\{ \displaystyle \min_f \sum_{i=1}^n V(f(\hat{x}_i), y_i) + \lambda \|f\|_{\mathcal{H}}^2 \right\}$$

3.0 Content Block

$$R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2^2 : w = \sum_{g=1}^G \bar{w}_g \right\} \quad \left\{ \displaystyle R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2^2 : w = \sum_{g=1}^G \bar{w}_g \right\} \right\}$$

4.0 Content Block

Proximal methods While the L_1 norm does not result in an NP-hard problem, the L_1 norm is convex but is not strictly differentiable due to the kink at $x = 0$. Subgradient methods which rely on the subderivative can be used to solve L_1 regularized learning problems. However, faster convergence can be achieved through proximal methods. For a problem $\min_{w \in H} F(w) + R(w)$ such that F is convex, continuous, differentiable, with Lipschitz continuous gradient (such as the least squares loss function), and R is convex, continuous, and proper, then the proximal method to solve the problem is as follows. First define the proximal operator

5.0 Content Block

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

6.0 Content Block

Other uses of regularization in statistics and machine learning Bayesian learning methods make use of a prior probability that (usually) gives lower probability to more complex models. Well-known model selection techniques include the Akaike information criterion (AIC), minimum description length (MDL), and the Bayesian information criterion (BIC). Alternative methods of controlling overfitting not involving regularization include cross-validation. Examples of applications of different methods of regularization to the linear model are:

7.0 Content Block

$R(f_1 \dots f_T) = \sum_{t,s=1, t \neq s}^T \|f_t - f_s\|_M^2$ for a given symmetric similarity matrix M .

8.0 Content Block

If W_{ij} encodes the result of some distance metric for points x_i and x_j , it is desirable that $f(x_i) \approx f(x_j)$. This regularizer captures this intuition, and is equivalent to:

9.0 Content Block

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

10.0 Content Block

Classification Empirical learning of classifiers (from a finite data set) is always an underdetermined problem, because it attempts to infer a function of any x given only examples $x_1, x_2, \dots, x_n$. A regularization term (or regularizer) $R(f)$ is added to a loss function:

11.0 Content Block

Group sparsity without overlaps Groups of features can be regularized by a sparsity constraint, which can be useful for expressing certain prior knowledge into an optimization problem. In the case of a linear model with non-overlapping known groups, a regularizer can be defined:

12.0 Content Block

Tikhonov-regularized least squares The learning problem with the least squares loss function and Tikhonov regularization can be solved analytically. Written in matrix form, the optimal w is the one for which the gradient of the loss function with respect to w is 0.

13.0 Content Block

$w_{k+1} = \text{prox}_{\gamma R}(w_k - \gamma \nabla F(w_k))$

14.0 Content Block

Without bounds on the complexity of the function space (formally, the reproducing kernel Hilbert space) available, a model will be learned that incurs zero loss on the surrogate empirical error. If measurements (e.g. of x_i) were made with noise, this model may suffer from overfitting and display poor expected error. Regularization introduces a penalty for exploring certain regions of the function space used to build the model, which can improve generalization.

15.0 Content Block

L1 regularization (also called LASSO) leads to sparse models by adding a penalty based on the absolute value of coefficients. L2 regularization (also called ridge regression) encourages smaller, more evenly distributed weights by adding a penalty based on the square of the coefficients.

16.0 Content Block

This can be viewed as inducing a regularizer over the L_2 norm over members of each group followed by an L_1 norm over groups. This can be solved by the proximal method, where the proximal operator is a block-wise soft-thresholding function:

17.0 Content Block

L_1 regularization can occasionally produce non-unique solutions. A simple example is provided in the figure when the space of possible solutions lies on a 45 degree line. This can be problematic for certain applications, and is overcome by combining L_1 with L_2 regularization in elastic net regularization, which takes the following form:

18.0 Content Block

Group sparsity with overlaps The algorithm described for group sparsity without overlaps can be applied to the case where groups do overlap, in certain situations. This will likely result in some groups with all zero elements, and other groups with some non-zero and some zero elements. If it is desired to preserve the group structure, a new regularizer can be defined:

19.0 Content Block

$\text{prox}_{\lambda, R, g}(\mathbf{w}_g) = \begin{cases} (1 - \lambda \|\mathbf{w}_g\|_2) \mathbf{w}_g, & \text{if } \|\mathbf{w}_g\|_2 > \lambda \\ 0, & \text{if } \|\mathbf{w}_g\|_2 \leq \lambda \end{cases}$